Ping of Death 공격 탐지 룰 생성

❶ [Kaliattacker] Ping of Death 공격

#hping3 --icmp --rand-source 192.168.10.30 -d 65000 -flood

```
root@kaliAttacker:~# hping3 --icmp --rand-source 192.168.10.30 -d 65000 --flood
HPING 192.168.10.30 (eth0 192.168.10.30): icmp mode set, 28 headers + 65000 data
 bytes
hping in flood mode, no replies will be shown
^C
--- 192.168.10.30 hping statistic ---
46882 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

- 문자열 65000 바이트로 네트워크에 ping전송

❷ [KaliAttacker-Wireshark] 공격패턴탐지

No.	Time	Source	Destination	Protocol	Length	Info
322900	13.56585200	192.168.10.20	126.219.151.120	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=19240, ID=e309)
322901	13.56585300	192.168.10.20	126.219.151.120	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=20720, ID=e309)
322902	13.56585500	192.168.10.20	126.219.151.120	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=22200, ID=e309)
322903	13.56585600	192.168.10.20	126.219.151.120	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=23680, ID=e309)
322904	13.56585700	192.168.10.20	126.219.151.120	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=25160, ID=e309)
322905	13.56585800	192.168.10.20	126.219.151.120	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=26640, ID=e309)
322906	13.56585900	192.168.10.20	126.219.151.120	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=28120, ID=e309)
322907	13.56586100	192.168.10.20	126.219.151.120	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=29600, ID=e309)
322908	13.56586200	192.168.10.20	126.219.151.120	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=31080, ID=e309)
322909	13.56586400	192.168.10.20	126.219.151.120	IPv4	1514	Fragmented IP protocol (proto=ICMP 1, off=32560, ID=e309)

```
0000  00 50 56 f2 e6 20 00 0c  29 62 9e 4e 08 00 45 00   .PV.. .. )b.N..E.
0010  05 dc e3 09 2d bb 40 01  83 4c c0 a8 0a 14 7e db   ....-.@. .L....~.
0020  97 78 58 58 58 58 58 58  58 58 58 58 58 58 58 58   .xXXXXXX XXXXXXXX
0030  58 58 58 58 58 58 58 58  58 58 58 58 58 58 58 58   XXXXXXXX XXXXXXXX
0040  58 58 58 58 58 58 58 58  58 58 58 58 58 58 58 58   XXXXXXXX XXXXXXXX
0050  58 58 58 58 58 58 58 58  58 58 58 58 58 58 58 58   XXXXXXXX XXXXXXXX
0060  58 58 58 58 58 58 58 58  58 58 58 58 58 58 58 58   XXXXXXXX XXXXXXXX
0070  58 58 58 58 58 58 58 58  58 58 58 58 58 58 58 58   XXXXXXXX XXXXXXXX
0080  58 58 58 58 58 58 58 58  58 58 58 58 58 58 58 58   XXXXXXXX XXXXXXXX
0090  58 58 58 58 58 58 58 58  58 58 58 58 58 58 58 58   XXXXXXXX XXXXXXXX
00a0  58 58 58 58 58 58 58 58  58 58 58 58 58 58 58 58   XXXXXXXX XXXXXXXX
00b0  58 58 58 58 58 58 58 58  58 58 58 58 58 58 58 58   XXXXXXXX XXXXXXXX
```

❸ 탐지 정책 생성

192.168.10.30으로 수신되는 '58585858'문자열을 가진 패킷이 10초안에 20번 들어오는 것을 탐지한다.

alert ip any any -> 192.168.10.30 any (msg:"Ping of Death"; content:"|58585858|"; threshold: type of both, track by_src, count 20, seconds 10; sid: 3000001;)

또는

송신되는 '58585858'문자열을 가진 패킷이 10초안에 20번 들어오는 것을 탐지한다.

alert ip any any -> any any (msg:"Ping of Death"; content:"|58585858|"; sid: 3000002;)

❹ [Kaliattacker] Ping of Death 재공격

#hping3 --icmp –rand-source 192.168.10.30 –d 65000 –flood

```
root@kaliAttacker:~# hping3 --icmp --rand-source 192.168.10.30 -d 65000 --flood
HPING 192.168.10.30 (eth0 192.168.10.30): icmp mode set, 28 headers + 65000 data
 bytes
hping in flood mode, no replies will be shown
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--- 192.168.10.30 hping statistic ---
46882 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

❺ [SecurityOnion-Sguil] 탐지결과 확인

ST	CNT	Sensor	Alert ID	Date/Time	Src IP	SPort	Dst IP	DPort	Pr	Event Message
RT	3	boan-virt...	3.38551	2017-09-11 14:41:50	192.168.10.10	43405	192.168.10.2	5815	6	ET SCAN Potential VNC S...
RT	5	boan-virt...	3.38552	2017-09-11 14:41:57	192.168.10.10	43405	192.168.10.20	5432	6	ET POLICY Suspicious in...
RT	10	boan-virt...	3.38566	2017-09-11 14:42:52	192.168.10.10	43405	192.168.10.200	1433	6	ET POLICY Suspicious in...
RT	9	boan-virt...	3.38572	2017-09-11 14:42:59	192.168.10.10	43405	192.168.10.200	1521	6	ET POLICY Suspicious in...
RT	1	boan-virt...	3.38618	2017-09-12 19:06:40	116.158.17.66	1489	192.168.10.50	80	6	ET DROP Spamhaus DR...
RT	1	boan-virt...	3.38619	2017-09-12 19:06:40	192.225.101.78	1737	192.168.10.50	80	6	ET DROP Spamhaus DR...
RT	1	boan-virt...	3.38620	2017-09-12 19:06:40	116.162.122.95	2180	192.168.10.50	80	6	ET DROP Spamhaus DR...
RT	1	boan-virt...	3.38621	2017-09-12 19:06:40	101.195.61.180	2577	192.168.10.50	80	6	ET DROP Spamhaus DR...
RT	1	boan-virt...	3.38622	2017-09-12 19:06:40	116.143.172.40	2587	192.168.10.50	80	6	ET DROP Spamhaus DR...
RT	2	boan-virt...	3.38662	2017-09-12 21:53:40	88.69.124.41		192.168.10.30		1	Ping of Death
RT	1	boan-virt...	3.38664	2017-09-12 21:53:40	153.61.10.129		192.168.10.30		1	Ping of Death
RT	2	boan-virt...	3.38665	2017-09-12 21:53:40	192.168.10.30		153.61.10.129		1	Ping of Death
RT	1	boan-virt...	3.38666	2017-09-12 21:53:40	209.132.167.162		192.168.10.30		1	Ping of Death

IP Resolution | Agent Status | Snort Statistics | System Ms

☑ Reverse DNS ☑ Enable External DNS

Src IP:
Src Name:
Dst IP:

☑ Show Packet Data ☑ Show Rule

alert ip any any -> any any (msg:"Ping of Death"; content:"|58585858|"; sid:1000006;)
/nsm/server_data/securityonion/rules/boan-virtual-machine-eth0-1/local.rules: Line 5

	Source IP	Dest IP	Ver	HL	TOS	len	ID	Flags	Offset	TTL	hkSu
IP	88.69.124.41	192.168.10.30	4	5	0	65028	159	0	0	64	566

	Type	Code	ChkSum	ID	Seq #
ICMP					